
[Course Title Here]

Master Summary

Stefan Eder

[Based on Lecture *name* SS26]

This document is a compact summary of the course. It aims to cover all relevant concepts and results, but long derivations, detailed proofs, and worked examples are often condensed or omitted in favour of clarity and conciseness. Use it as a study aid alongside the lecture notes and slides, not as a replacement for them.

If you spot an error or a gap, please open an issue or pull request at github.com/steveTgit/AI_Master — corrections are always welcome.

Good study materials should be shared, not gatekept. If this helped you, take it as a reason to help the next colleague you see struggling with something.

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Example Chapter Title

This is the introductory running text for the chapter. Chapters always begin on a new page. Use this space to frame the chapter's goal at a high level before descending into sections.

1.1 First Main Concept

Lead every section with prose. Introduce the idea, its motivation, and any notation before defining it formally in a box. Inline math flows naturally: a model parameterized by $\boldsymbol{\theta} \in \mathbb{R}^d$ maps an input $\mathbf{x}$ to a prediction via $f_{\boldsymbol{\theta}}(\mathbf{x})$. Display math is used when an expression needs space:

$$\mathcal{L}(\boldsymbol{\theta}) := -\frac{1}{N} \sum_{i=1}^N \log p_{\boldsymbol{\theta}}(y_i | \mathbf{x}_i).$$

Cross-Entropy Loss

For a model with parameters $\boldsymbol{\theta}$ trained on $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, the *cross-entropy loss* is:

$$\mathcal{L}(\boldsymbol{\theta}) := -\frac{1}{N} \sum_{i=1}^N \log p_{\boldsymbol{\theta}}(y_i | \mathbf{x}_i).$$

After a definition box, always continue with prose. Here you would explain what this loss measures, why minimizing it corresponds to maximum likelihood estimation, and how it connects to what comes next.

1.1.1 A Deeper Dive

Bayes' Theorem

$$p(\boldsymbol{\theta} | \mathcal{D}) = \frac{p(\mathcal{D} | \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathcal{D})} \propto p(\mathcal{D} | \boldsymbol{\theta}) p(\boldsymbol{\theta}).$$

The posterior is proportional to the likelihood times the prior.

Note how the theorem box below does not immediately follow a section heading — there is always at least a sentence of prose first to give it context.

1.2 Practical Applications

Running text sets up the example and the intuition before those boxes appear. Consider what the KL divergence $D_{\text{KL}}(q \parallel p)$ actually measures.

Intuition & Mental Model

Think of $D_{\text{KL}}(q \parallel p)$ not as a symmetric distance (it isn't — $D_{\text{KL}}(q \parallel p) \neq D_{\text{KL}}(p \parallel q)$ in general), but as the *extra bits* needed to encode samples from q using a code optimized for p . The asymmetry is exactly why direction matters in variational inference and the EM algorithm.

Simple Integration

Let $f(x) = 2x$. Compute $\int_0^2 f(x) \, dx$:

$$\int_0^2 2x \, dx = [x^2]_0^2 = 4 - 0 = 4.$$

Exam Trap

Do *not* confuse $D_{\text{KL}}(p \parallel q)$ with $D_{\text{KL}}(q \parallel p)$. In variational inference we minimize $D_{\text{KL}}(q \parallel p_\theta)$ (forward KL $\rightarrow$ mass-covering) vs. $D_{\text{KL}}(p_\theta \parallel q)$ (reverse KL $\rightarrow$ mode-seeking). The behavior of the approximation differs dramatically.